

CHAPTER 2

HOW BOUNDARIES ARE CREATED

2.1 INTRODUCTION

To understand the legal fiction of boundaries, the surveyor must be able to make a distinction between the various elements of which boundaries are composed. In most instances, but not all, a boundary must have a corner at each end of the invisible boundary line. A monument may be placed at each corner point, or the corner may exist by reference to a document. The boundary line to be identified by the surveyor in conducting his or her retracement may be a deed line, a possession line, or a claim of ownership line. The deed line is factual the possession line is legal, and the claim line is mental. Corners, monuments, boundary lines, and property lines should all mean different things to the surveyor; they should be addressed differently by the surveyor who creates them than by the surveyor who must ultimately retrace them and who may redescribe them in subsequent documents and then probably will have to defend their location in court. Unfortunately, these principles are sometimes misunderstood by students, practicing boundary surveyors, attorneys, and even the courts.

A deed line is factual; a property line is legal.

Neither the student nor the surveyor nor the court should confuse the distinction between the creation of existing boundaries and changing boundaries already in existence and identifying the location of a boundary that is to be retraced. The changing of boundary lines already created will be discussed later in this book; for the most part, the principles are legal in nature.

The following principles are discussed in this chapter and, in most instances, are addressed by attorneys, but they should be understood by the courts and especially by the retracing surveyor.

- PRINCIPLE 1. A landowner may divide a parcel of land in any manner not inconsistent with the law.
- PRINCIPLE 2. There can be only one original survey that creates the boundaries of a parcel.
- PRINCIPLE 3. The “footsteps” of the original survey are the evidence left on the ground as well as the documentation identifying the ground evidence that the original surveyor created.
- PRINCIPLE 4. The retracing surveyor must be able to locate and then make a nexus of the ground evidence to the written evidence.
- PRINCIPLE 5. Once a boundary or boundaries are created, no alterations or modifications are permitted in any manner by either the landowner or any surveyor once property rights have been granted or distributed according to the boundaries created.
- PRINCIPLE 6. The original surveyor creates boundaries. It is the retracing surveyor who ascertains or identifies boundaries from the original evidence.
- PRINCIPLE 7. Once created and approved, the original boundaries created are legally without error and are the exact dimensions as indicated by the creating surveyor.
- PRINCIPLE 8. No surveyor or court has the legal authority to alter or modify a boundary line once it is created. It can be interpreted only from the evidence of where that boundary is located based on the “footsteps” in Principle 3.
- PRINCIPLE 9. A retracing surveyor relates previously created and recovered evidence to a current survey, being mindful that the current survey is always subject to collateral attack by other surveyors.
- PRINCIPLE 10. A modern survey may describe bearings and distances to a greater degree of precision using modern technology, but it can be no more precise than the original measurements that created the original lines.
- PRINCIPLE 11. When there is a scarcity of written evidence that indicates no lines were created on the ground, any survey conducted subsequently may be held as the first survey. The courts may consider the first survey to be the original survey, or they may consider the original survey to be the first survey.

2.2 DEFINITIONS

Several books give academic and/or legal definitions of words that are important to the practicing surveyor. These are the definitions that should be used if one were

to testify in court. But in many instances they fail to meet the everyday practitioner's requirement for definitions that are understood. In an effort to introduce a degree of practicality, the definitions given here are those that the practicing surveyor, student, and landowner can appreciate without legal or academic confusion.

Any list of definitions can be short or extensive. The definitions presented here are only those basic ones that are needed to furnish a foundation for the understanding and the study of boundaries. Corners, lines, monuments, property lines, and boundary lines may have several definitions. The following are those that the authors use to make distinctions.

A *line* in surveying, and which divides interests in real property, means, *prima facie*, a mathematical line [invisible] without breadth; however, this theoretical idea of a line may be explained, by the facts referred to and connected with the division, to mean a wall, a ditch, a crooked fence, or a hedge—that is, a line having breadth.

Usually, a *corner* is at the end of a boundary line or at a change in direction of a boundary line. Applying this concept, we see why an endpoint is also called the *terminal point* of a line, named after Terminus, a deity to the ancient Greeks and Romans. A corner may also be placed along a line where a third party may tie in or reference a senior line. To define a line, there must be two corners, or termini, one at each end of the line. The corners, to be controlling over other descriptive elements found in descriptions, must be created by the survey and must be called for in a legal document relating to the specific parcel.

In many instances, corners may be referenced in a description as *points*. Historically, points were described in early surveying textbooks. As early as 1657, William Leybourn referenced a point as being “without unity.”¹ Then, in 1687, John Love, in *Geodaesia*, referred to a point as “that hath neither Length nor Breadth, the least thing which can be imagined, and which cannot be divided. . . .”²

Not until some person places a physical object, the monument, at the *point* does the point become identifiable and retracable.

Corners can be located or placed on the exterior boundaries of the parent parcel as well as inside the parcel itself. Usually, an interior corner will not control an exterior corner. To be controlling and legal, a corner does not have to be monumented. A corner has no physical dimensions of length, width, or depth. It has only a legal dimension, in that an original corner found legally identifies the point of the survey, regardless of whether it is monumented. This applies only if a conflict exists in a description; otherwise, it can control even if it is not called for, but this is predicated on other applications of law. When in a retracement a corner that was originally described as a *point* is identified, with no reference to having been monumented, it is difficult, if not impossible to identify with certainty in a subsequent survey. The original “point” was a product of the equipment and methodology used at that time, and these two elements cannot be totally replicated. The placement of that original “point” is unique. The certainty of identification and position become recoverable only when the original point is monumented and then reduced to writing.

To be legally controlling, a *monument* is a physical manifestation set at or near a corner. A corner does not have to be monumented; it just has to be identified, with clarity, and called for in the survey or document. However, the law has set certain

requirements for corners. As we will see, a call for a distance, more or less without the call for a corner or monument, is legally insufficient. Yet we can have monuments placed on the corner point or near the corner either as a substitute for the corner point or to define possible multiple locations of the same point. Surveyors and landowners have problems when a monument is not placed at the corner point but near the point where the corner is located. One of the critical requirements for a monument to be controlling is that the *monument* recovered in a retracement be called for by words and the recovery description, not differing appreciably from the monument set at the time of the original survey, when the corner was established and a monument was set to identify that specific point.

Monuments may be classified as falling into two areas or descriptive categories: natural and artificial. Courts usually make a legal distinction between the two and hold that when there is a conflict between or question about a natural monument and an artificial monument, the natural monument usually will control legally.

Courts have held that streams, rivers, mountains, roads, and trees are supposed to identify a point that represents a corner—the legal point. But how can a stream that is 20 feet wide and 600 feet long be considered a monument? This discrepancy cannot be explained. It is even more difficult to explain how you can have a boundary line without having terminal corners. The reader will find that this is but one of many possible conflicts that may be encountered by the professional surveyor in retracement work. Research of early U.S. case law reveals that many of the cases called these points *natural boundaries* and *artificial boundaries*.

Artificial monuments are those monuments that usually are placed at corner points by landowners, surveyors, engineers, and others. They may be referred to by various terms on maps, in descriptions, and in field books. One may find them referenced as iron pins or pipes (IPs), stakes, and, at times, trees, concrete monuments (CMs), nails, or whatever the surveyor decided to use or was available from the back of the survey truck. Courts have made the distinction that natural monuments control over artificial monuments, because their identification is more certain and they are less likely to be disturbed. (This is discussed in Section 5.10.)

The surveyor must make the distinction between a *boundary line* and a *property line*. Boundary lines between parcels are created in several ways, yet until written documents or legal principles attach, property lines are nonexistent. In theory, a boundary line remains fixed forever where it was located initially, but a property line may change by means of legal principles, including estoppel, agreement, adverse possession, or riparian rights.

When a surveyor establishes a survey line in the field to create a boundary, corners are usually set and then described and identified. Lines are then marked and monuments set at the corners. By principle, each line must have two corners, one on each end, to be fixed in place. Although, when placed, the monuments become visible to the surveyor and landowners, and the monuments that are set identify the position of the corners, the connecting lines between the corners remain invisible. However, if the creating surveyor blazed trees, or placed posts or stakes along that line, these physical objects become the identification of that once-invisible line. Thus, the objects marked *at the time of the survey* become the location and

identification of the boundary line created. The recovery of any of the evidence of this survey at a later date becomes the recovery of the original line, and the lines run in the field to create the original boundary controls.

Once a boundary line is created in the field or on paper, common law and statute law permit it to be modified, changed, and altered. When a line is thus changed, altered, or modified, according to law, it becomes a property line. Because surveyors can create boundary lines and because a property line is a result of law, only the courts can certify or “sanctify” property lines or boundaries. Surveyors in the United States have not been given the authority to determine legal property lines. Usually, when two surveyors disagree on the survey location of a line, the parties resort to the legal system to make the final determination. This is often referred to as “the battle of the surveyors.”

Once a boundary line is created and identified, it maintains its legality and position even if the corners on its extremities become lost. This may be difficult to understand, but the law recognizes a boundary line as being a legal entity, and, once it is created, as long as evidence of that line exists, the line is controlling.

On the other hand, although a true boundary is controlling, surveyors, in retracing these true boundaries, may consider it necessary and prudent to evaluate property boundaries. In many instances, the true boundary and the property boundary are one and the same; that is, they are superimposed on each other. The fence line or hedge line between the two adjacent parcels is superimposed on the deed line that was surveyed on the ground. However, the common law and statute law of most states permit people to modify these deed boundaries through legal doctrines, some of which are centuries old. Although they may be referred to by different terms in different states, for the sake of identification we will call them *agreement*, *estoppel*, *acquiescence*, and *adverse possession*. These doctrines require a landowner to complete some act in order to perfect the change of the boundary. Changing a boundary through a riparian change must be through, or a result of, an act of nature.

2.3 CLASSIFICATION OF BOUNDARIES

When a surveyor or an attorney looks at a boundary, he or she usually does not consider the magnitude or the complexity of that boundary. One may be asked either to create or to retrace a boundary of an entire parcel, a boundary of a portion of a larger parcel, or the entire boundary of a single minute parcel. The boundary may be composed of natural elements, artificial elements, words only, or a combination of the three. It may be either a complete boundary or a portion of a larger boundary.

These *macro* and *micro boundaries* enjoy the same attributes of definition in that they commence at a common point in the center of the Earth and extend to the accolades in the heavens. This was tested when the former Soviet Union made a claim extending its boundaries to the North Pole, at a distance of 20,000 feet under the Arctic Ocean. As an afterthought: When macro boundaries are disputed, nations bring out the nuclear weapons; when micro boundaries are disputed, the individuals involved resort to guns, knives, and fists.

As discussed in Chapter 1—History and Concept of Boundaries, all boundaries may be considered as either macro boundaries or micro boundaries. The distinction is not necessarily the manner in which the boundary was created but the rights or interests that were being separated and identified.

Since *macro boundaries* usually separate major interests, boundaries between nations, boundaries between political subdivisions of a nation, and major private or governmental subdivisions may be placed in that category. *Micro boundaries* include smaller subdivisions within macro boundaries. Using this distinction, a boundary may be a macro as well as a micro boundary.

A township and range within a state may be a micro boundary within the state, but in relation to the respective sections identified within its boundaries, the township is a macro boundary for the respective sections (Section 3.2).

The law of retracements may apply differently to township lines (macro boundaries) than it does to section lines (micro boundaries). Consequently, if a subdivision of homes (micro boundaries) is placed within the boundaries of the section (macro boundaries), entirely new rules of retracement apply. When there are conflicts between boundaries, the problem must be approached in an orderly and systematic manner, using logic, analysis, and experience to arrive at a possible and suitable solution.

Having classified boundaries as macro or micro, they may be further classified as to the authority for their creation into two broad categories:

1. Rectangular
 - a. Created by federal statutes
 1. Rectangular
 2. Metes and bounds
 - b. Created by state statutes
 1. Rectangular
 2. Metes and bounds
 - c. Created by private entities
2. Metes and bounds

Regardless of how the original boundaries were created or what category the retracing surveyor may place them in, the proper rules of retracements should be followed.

These categories are discussed in detail in later chapters. It will suffice at this time to recognize that the various methods of creation will help practicing surveyors understand that the methods and laws by which boundaries are created affect the methods of retracement.

2.4 METHODS OF BOUNDARY CREATION

In acknowledging the fact that a boundary is a factual invisible division line between two interests or estates in land, one must look at both common law and statute law

to surmise how courts and legislative bodies have and will determine the various methods by which boundaries can be created. In applying the methods of boundary location, the surveyor can identify multiple methods by which the surveyors, the courts, or a landowner can create a boundary. It should be remembered that one does not have to be a surveyor to create a boundary. Courts can adjudicate and dictate a boundary, two or more contiguous landowners can create mutual boundaries, a single landowner can create a boundary by personal actions or words, and surveyors can create boundaries by their actions or words.

For study purposes, boundaries can be determined or created as follows:

1. *By action.* This is the actual creation of a boundary on the ground by survey, using the instrumentation and the words that are current at that time, by individuals or by actions. Corners are created and identified, monuments established, and notes made of the survey or actions; possibly, a plat will be prepared and lines marked on the ground. These actions are then incorporated into the resulting descriptions.

2. *By words.* Exterior and interior boundaries of tracts may be created by words. A deed describing “the south ¼” or “the north 10 acres” creates boundaries, as does a metes and bounds description having courses (bearings and distances) and corners. A major problem is that a “south ¼” description has no calls for monumented corners. If the description is a result of a survey, original lines are created and the surveyor is an original surveyor. If the description is a words-only document, there is no original surveyor, only a first surveyor who locates the lines described from the words; as such, this survey is always subject to future collateral attack.

3. *By law.*

- a. *Statute law.* Boundaries can be a result of statutes. These boundaries may also be identified under point 1. The difference is that, whereas under method 1 the actions are performed as a result of a mutual desire to locate the boundary, under this section a governmental body, either federal or state, enacts legislation duly authorizing certain surveyors to survey large tracts of federal- or state-owned lands.

The most noted and extensive surveys were those conducted by the federal government, now known as the Public Land Survey System (PLSS) or General Land Office (GLO) system. This system was created when several states, Georgia and Texas in particular, enacted legislation directing how their public lands would be surveyed for disposal to their citizens. Although the state legislation was not as extensive or as detailed as the federal legislation, it did identify such areas as how the respective parcels would be surveyed and the type of equipment to be used by the surveyors. More particularly, it directed how they would be identified and distributed. Some also included the qualifications that were required of surveyors. The federal laws that created the public land system (GLO) were supplemented with a subsequent act (Land Act of February 11, 1805) that basically set the dignity of the original surveys.

- b. *Common law.* Although the common law has recognized that individual landowners can, under certain situations, establish their own unilateral or mutual boundaries, many states have identified, by statute, the specific requirements necessary to meet these historical methods.

Accepting the fact that the laws recognize the creation of such boundaries, the names often become confused, misapplied, misidentified, and generally muddled by landowners, attorneys, and, especially, the courts. In the founding days of our country, many lawyers and judges were also schooled in surveying, title examination, and land principles. This cannot be said of many of the younger professionals today. Not to detract from their capabilities, but today's young legal professionals have not been schooled in basic land principles; they usually lack a basic "land ethic." On occasion, the principles of agreement, estoppel, acquiescence, and the age-old doctrine of adverse possession to change, alter, and reidentify boundaries have to be identified, understood, and then accepted.

Creation of Boundaries by Running Lines

The foremost method of boundary creation, and the one best understood by the courts, is that of running on the ground by the surveyors and then reducing this running to words by creating field notes and plats, and possibly descriptions.

It is well recognized by common law and case law that a person may create boundaries of parcels, lots, and even subdivisions by their actions. A basic underlying principle is that as long as the person does not go beyond the exterior boundaries of the title lines, a parcel may be subdivided in any manner the law permits by running and establishing lines on the ground. This is not in contradiction to establishment by law, but it could be considered as establishment by common-law principles.

In the past, landowners were permitted to subdivide their parcels into lots of any size without any restrictions. Today, numerous subdivision laws have been enacted that limit the size of parcels and their potential uses. However, the retracing surveyor should realize that this does not resolve the issue of those many parcels that were created and sold with no restrictions—in many instances, the subdivision plats were not made a matter of public knowledge by registration.

If a person owns a parcel and wishes to subdivide it into smaller segments, this can be done by physically running survey lines, creating corners, either setting or not setting monuments, marking lines, creating field notes, and possibly preparing a plat of the subdivision. The landowner then conveys the individual lots as they were created. The description by which the lots were conveyed may or may not refer to the survey. If it does, the true intent is identified; that is, the survey would control the lines, regardless of any latent errors.

If a survey is not referred to, the rule is not clear. Courts have been inconsistent: Some courts have permitted reliance only on circumstantial evidence that the original survey controls; others have relied on direct evidence. Some have held that if a plat is part of a description, all elements on the plat become part of that description, including the survey that created it, which includes the description of the lines run and the monuments set. Others have permitted surveyors to testify as to what was done and made their testimony controlling. To understand what local courts permit, each surveyor should become familiar with local legal decisions regarding boundaries as well as accepted universal legal principles.

Creation of Boundaries by Verbal Actions

If we were to consider maps, plats, and descriptions as verbal actions, it is well recognized at law that a landowner can create a boundary without benefit of survey. This may also include actions by landowners walking out lines, establishing corners, and setting monuments and then writing their own deeds and descriptions to which rights will be conveyed.

Many times, a surveyor will be asked to locate a parcel described as “north ½,” “east ¼,” or the “south 10 acres” from a description prepared by the landowner. In analyzing such descriptions, there is one certain principle that surveyors, attorneys, and the courts must consider: In some states it is not unlawful for a landowner to convey a portion of the parent parcel by a description without the benefit of a survey. This occurs when a person owns a 10-acre parcel and conveys out the “south 5 acres” or the “south ½.” These two descriptions may or may not be inconsistent. Yet once the deed is delivered or signed, an invisible boundary is created between the “north 5 acres and the south 5 acres” or the “north ½ and the south ½,” even though no survey was ever conducted. The responsibility of the surveyor is to be able to place this invisible line on the ground by survey. This resulting survey is not an original survey that a surveyor creates when he or she runs the line first before making the plat. It should be considered as the first survey. The unique situation this first survey is placed in is that it is not controlling, as the original survey would be. A first survey by the initial surveyor is no more controlling than would be any other subsequent survey. This first survey is but an interpretation or an opinion by the surveyor or surveyors of the written description from which the surveyor works.

Creation of Boundaries by Survey

The attributes necessary for the surveyor who is given the responsibility of creating boundaries is entirely different from the attributes of the surveyor who is given the responsibility of finding the boundaries originally created or when conducting a subsequent retracement. We do not have the flexibility to pick and choose our surveyors for certain jobs, for registration laws hold that, once a surveyor is registered, that surveyor is legally and professionally capable of performing any of the actions that the law permits. *Being legally capable is not the same as being technically capable.* The surveyor who undertakes the responsibility either to create or to retrace boundaries will be assumed to be totally qualified to conduct either job once it is started.

Failure to follow the basic survey principles in creating the original surveys and the basic principles in conducting retracements may be tantamount to *negligence per se*.³

2.5 WHO MAY CREATE BOUNDARIES?

Principle 1. *A landowner may divide a parcel of land in any manner not inconsistent with the law.*

Notwithstanding laws to the contrary, landowners may subdivide land in any manner to suit their needs and desires. If questioned, it may well be considered a constitutional right to do so. Under the common law, one of the benefits of landownership is the ability of a landowner to sell property and describe this property in any manner the property owner wishes. At common law, there are no requirements that a landowner conduct a survey as a prerequisite for conveying a parcel of land or a property interest. There are distinctions as to the type of description that may be legal as far as the courts are concerned and adequate as far as a surveyor is concerned.

In many instances, when the landowner conducts his or her own survey or creates his or her own description and deed, a substantial monetary savings to the parties usually results, only to be offset later by the possible costs of legal fees and litigation between the original parties or future parties in interest. Surveyors should discourage and not aid landowners who practice this authorized but not recommended activity.

The property owner, unless prohibited or restricted by law, is empowered with the authority to subdivide the area in any manner he or she desires. Usually, there are no restrictions on how the original lines are marked or described.

Unknowningly, some landowners create a future opportunity for confusion and possible litigation by changing a boundary description from one method to another. Land that is created by an aliquot description will be redescribed to another form. For example, an original description that says “the S $\frac{1}{2}$ SE $\frac{1}{4}$ of Section 7” will be redescribed as “the South 80 acres” or it may even be presented as a metes and bounds description with or without the call for monuments.

A landowner may even take a large parcel of land and create his or her own rectangular system within the parcel.

The retracing surveyor must be “tuned in” when this happens and properly apply the appropriate retracement techniques and rules of descriptions.

Principle 2. *There can be only one original survey that creates the boundaries of a parcel.*

Individual surveyors and attorneys, as well as judges, confuse what an original survey actually is. When having to decide whether the survey being retraced is an original survey, the retracing surveyor should look at the original titles to the property. The original survey should be correlated to the boundaries described in the original title. Because of the acceptance of the doctrines of estoppel, agreement and acquiescence possession lines may affect the written deed boundaries.

Once approval and title(s) have been granted in reference to that survey, the creator of the parcels cannot alter any boundaries created without the approval of all interested parties.

When boundaries are referenced to one survey and the titles to the parcels are referenced and conveyed to the described boundaries, they become unalterable without legal action.

Principle 3. *The “footsteps” of the original survey are the evidence left on the ground as well as the documentation identifying the ground evidence that the original surveyor created.*

Evidence of the original boundaries can be misidentified by professionals who must conduct retracements with the hope of identifying the original locations of the originally created lines and the resultant established corners.

In order to conduct an adequate retracement, courts have held that, to have the totality of the evidence, the surveyor conducting the research should consider that the instructions given to the creating surveyors, the fieldwork, the trees marked, the monuments set, the plats prepared, and any reports created are all part of the original survey and should be obtained and considered.

Principle 4. *The retracing surveyor must be able to locate and then make to a nexus of the ground evidence to the written evidence.*

A found monument, supposedly placed by some unknown person at some unknown time, should not be used as representing a described corner in order to be used in a retracement or to prove the original location of a corner. This can be said about any evidence the retracing surveyor wants to use to prove the existence of any survey, including an original survey. The retracing surveyor must have the documentary evidence that described the corners and its monument in order to consider the monument as being a controlling element for one termination of a boundary line.

A found monument with no written history of its placement has no legal or survey value indicating the boundary.

Some surveyors and courts accept found monuments as primary evidence to solve boundary issues, but a found monument without a historical written connection is of little value without some other species of legal and survey support. In the novel *Uncle Tom's Cabin*, Little Topsy was asked the question; "Topsy, just where did you come from?" Her response was "I don't know. I just grewed." If you are unable to determine a found monument's origin, it is better not to use it as a controlling element in your survey.

Principle 5. *Once a boundary or boundaries are created, no alterations or modifications are permitted in any manner by either the landowner or any surveyor once property rights have been granted or distributed according to the boundaries created.*

This principle may cause confusion and subsequent problems for landowners, surveyors, and attorneys, and possibly neighboring landowners. One of the guarantees of the U.S. Constitution is that of being able to use land without government control or intervention. In modern times, that has been modified by zoning laws, ordinances, covenants, and so on. But one may survey and convey property in any manner not inconsistent with the law. This permits the landowner to survey and create boundaries for conveyancing. However, under the law, once these boundaries are created, the landowner can modify the boundaries in any manner desired as long as

no property rights have been conveyed to third parties. But once a single lot or interest is conveyed according to these micro boundaries, all micro boundaries within that macro boundary become legally fixed and cannot be altered or modified without approval of all persons who have vested property rights.⁴

This is a difficult concept for landowners to comprehend while they still own the parent parcel, but it should be understood that the conveyance of a single lot in reference to the plat or subdivision line “seals” the location of all lines and corners referenced and identified on the plat. (Not recording the plat will not affect this principle.)

At common law, there are no restrictions as to how the landowner can convey these lands. Choices must be made as to whether a survey is to be made to identify the lines or the landowner is simply to convey the land using words.

Principle 6. *The original surveyor creates boundaries. It is the retracing surveyor who ascertains or identifies boundaries from the original evidence.*

In keeping with Principle 1, the landowner can create or establish boundaries on his or her own property, and the survey is the method by which these boundaries are actually created on the ground. A parcel of raw land has no boundaries. But once the surveyor runs and then identifies these lines, the boundaries are created and can never be altered by any subsequent surveyor.⁵ Once these created lines are used to identify parcels. The elements of the original and all related information created from or identified from the original survey cannot be altered in subsequent surveys. The distances recited are those distances indicated; a subsequent surveyor only retraces the original survey, and the remeasured lines are only redescribed in relation to more modern and possibly more precise measurement(s). Because boundaries are created, the original actions cannot be altered by subsequent surveyors or landowners. It is subsequent surveyors who ascertain the location of the original boundaries by conducting retracements⁶ based on the recovery of the original evidence and then relating this evidence to the description of the original boundaries.

The subsequent surveyor conducts a retracement that differs from an original survey both legally and in terms of permanency. Whereas the law has deemed the original corners established by the original survey to be legally errorless and unalterable, the subsequent retracement has no permanency. This was discussed in the Florida decision *Rivers v. Lozeau*.⁷

. . . [A] surveyor can be retained to locate on the ground a boundary line which has theretofore been established. When he does this, he “traces the footsteps” of the “original surveyor” in locating existing boundaries. Correctly stated, this is a “retracement” survey, not a resurvey, and in performing this function, the second and each succeeding surveyor is a “following” or “tracing” surveyor and his sole duty, function and power is to locate on the ground the boundaries, corners and boundary line or lines established by the original survey; he cannot establish a new corner or new line terminal point, nor may he correct errors of the original surveyor. *He must only track the footsteps of the original surveyor. The following surveyor, rather than being the creator of the boundary line, is only its discoverer and is only that when he correctly locates it.* (Emphasis added.)

2.6 SANCTITY OF THE ORIGINAL SURVEY

Principle 7. *Once created and approved, the original boundaries created are legally without error and are the exact dimensions as indicated by the creating surveyor.*

Although this principle is codified so far as federal lands are concerned in federal law in the Land Act of February 11, 1805, it is also accepted in common law for the metes and bounds states. When it becomes necessary to understand the full power of the original lines, it then becomes important for the creating surveyor, the retracing surveyor, and the courts to understand and appreciate the dignity of a found, identified original survey. The creating surveyor should strive to use the most precise means to run the surveys that create the boundary lines as accurately as possible and then record the actions definitively. The retracing surveyor should realize that modern methods of measurement will not duplicate the measurements that created the original boundaries. The courts must then realize that these original lines cannot be altered by judicial determination in that they are unassailable through any court, including the U.S. Supreme Court. The retracing surveyor, as part of the totality of the evidence that is recovered, then should correlate his or her retracement measurements to the original measurements. A problem may present itself when the surveyor's research indicates that the original measurements are in one unit and his or her remeasurements are in a totally different unit (e.g., the original description is in poles and the remeasurements are in feet).

Historical correlation of measurements, angles, and distances in full degrees and poles and retraced distances should be in the same units.

The responsibility of surveyors is to examine evidence that is presented to determine where the true and correct locations of these lines are located. When attempting to locate these original lines, subsequent surveyors should always make their analyses in terms of the original units of measurements.

2.7 ORIGINAL LINES REMAIN FIXED

Principle 8. *No surveyor or court has the legal authority to alter or modify a boundary line once it is created. It can be interpreted only from the evidence of where that boundary is located. It can be interpreted only from the evidence of where that boundary is located based on the “footsteps” in Principle 3.*

This basic principle was created by the U.S. Supreme Court in the decision *Cragin v. Powell*⁸ when it wrote that original surveys are *unassailable through the courts*.

Retracing surveyors will encounter a minority of surveyors who, when finding an “error” in the original survey, believe it is their responsibility to “correct” the

error and make the original bearings and distances as they should be had they been surveyed correctly. These surveyors have no concept that, once the lines have been created, no subsequent surveyor has authority to re-create the original lines. When a creating surveyor indicates a distance or an angle, these are the original measurements, according to the creating surveyor's methodology and errors. By law, they are free of error, even though, in fact, we do realize that the creating surveyor made mistakes. This concept also extends to the presumption that the survey is correct⁹ and that no subsequent surveyor or surveyors can correct the original lines; they can only redefine them. It is not the job or responsibility of subsequent surveyors to correct the originals. It is their job to report any discrepancies found. Differences do occur because of the different methods and equipment used in obtaining the original measurements and the subsequent measurements, as well as differences among the people who did the work and changes in the circumstances or conditions under which it was done.

2.8 DISTINCTIONS BETWEEN THE ORIGINAL BOUNDARY SURVEY, THE RETRACEMENT SURVEY, AND THE FIRST SURVEY

People, including many surveyors and attorneys, use words freely. When a person needs a word to explain a situation or any evidence and a word is not readily available, they are likely to create a name, which may become fixed in its use. This relates to these three terms:

Original survey

Retracement

First survey

Such words as *love* and *friend*, as used in everyday speech, often have meanings that are different from their true meanings. Surveyors use the term *property boundary* very freely on maps, in reports, and in everyday conversation, yet no surveyor has the authority to identify and locate property boundaries, and few states define one of the attributes of surveying as "locating property boundaries." Without judicial authority, no surveyor has authority to locate or identify property boundaries.¹⁰ Interestingly, though, several states have now added, as a distinct category of surveying, giving testimony in court as to boundaries.

Principle 9. *A retracing surveyor relates previously created and recovered evidence to a current survey, being mindful that the current survey is always subject to collateral attack by other surveyors.*

Having seen that only the original entity that created the original boundaries can change boundaries after having secured the sanctity of vested property rights, the surveyor in the private sector can only conduct a retracement of an original survey.

The concept of the survey is best defined in *The Manual of Instructions*.¹¹ The *Manual* defines the areas of resurveys and retracements in the following areas (see Chapter 1—History and Concept of Boundaries):

1. Original surveys
 - a. Retracements
 - b. First survey
2. Resurveys
 - a. Dependent
 - b. Independent

Since an original survey may be conducted of both public and private lands, both independent and dependent resurveys may be conducted only by the entity that conducted the original survey. On the other hand, a retracement is a survey that is made to ascertain or redefine the direction and length of lines and to identify monuments that were established at corner positions by a prior original survey. A retracement is usually conducted within a macro boundary (e.g., a township) to ascertain micro boundaries (e.g., sections). It may be necessary to retrace several miles of lines within a macro boundary from found, proven, original corners to set or relocate a lost corner or corners on a micro boundary.

A retracement may be used to recover new and additional evidence to ascertain the quality of an earlier survey. In a retracement, survey-found proven corners are rehabilitated, but any lost corners that are resurveyed have no finality of position.

The place of a retracement (resurvey) is best explained by the following excerpt from *Cragin v. Powell*:¹²

The making of resurveys or corrective surveys of townships once proclaimed for sale is always at the hazard of interfering with private rights, and thereby introducing new complications. A resurvey, properly considered, is but a retracing, with a view to determine and establish lines and boundaries of an original survey . . . but the principle of retracing has been frequently departed from, where a resurvey (so called) has been made and new lines and boundaries have often been introduced, mischievously conflicting with the old, and thereby affecting the areas of tracts which the United States had previously sold and otherwise disposed of.

In the first survey, lines are difficult to ascertain, but in most instances they do not control as do the original lines.

2.9 ORIGINAL TECHNOLOGICAL METHODS OF BOUNDARY CREATION NOT RELATABLE TO MODERN METHODS

Principle 10. *A modern survey may describe bearings and distances to a greater degree of precision using modern technology, but it can be no more precise than the original measurements that created the original lines.*

In attempting to analyze a historic description of a boundary for retracing, an analysis should be made that compares the methods and equipment used for those that created the boundaries to the methodology that will be used to take the historical description and retrace that description today using the most modern of surveying equipment.

In many instances, ancient bearings and distances are the references for the modern survey. Although the more modern equipment may be more precise than the original equipment, the newly retraced boundaries can never replace the accuracy of the original surveys. Today, the surveyor is asked to retrace bearings and distances that may have been created hundreds of years ago and then relate them to references that are no longer used in descriptions. Until recent years, the universal instrument used to create bearings and their companion instances were the compass and the 2-pole chain, consisting of 50 links totaling 33 feet. A compass initially was graduated to a full degree with an estimated reading of $\frac{1}{4}$ of a degree (*not* 15 minutes). Usually, distances were measured with a 2-pole chain consisting of 50 links of 0.66 feet. The individual link could be estimated to probably 0.2 of a link (*not* 0.132 feet).

Today, we take these as absolutes and make a conversion into absolute reality, and then, using these absolute conversions, we use modern equipment to place on the ground these inexact references. In the process, we fail to consider the significant number relationships in the conversions.

If the retracing were to take the converted bearings and distances as absolutes, determining relationships would be made difficult, if not impossible.

2.10 ORIGINAL LINES MAY BE REDESCRIBED AS A RESULT OF A RETRACEMENT

Applying this principle with wisdom and foresight, American jurisprudence (the courts) anticipated that technology would probably change in the future, realizing that there has to be a finality to measurements because technology has refined the original measurements more precisely.

The courts have made the distinction between accuracy and precision. The major decisions held that the original measurements were without error in the location of the monuments, when found undisturbed, and thus they control. The basic premise is that the angles and distances cited in the field notes and then identified on the plats are the exact measurements created by the original surveyor using the equipment and methodology of that time, but any subsequent “survey” will use different equipment and different personnel for the subsequent work, and thus it is not and cannot be a duplication of the original work.

All measurements have a positional tolerance of location at the endpoints.

2.11 CONCLUSIONS

In application of this chapter to students and practicing surveyors, as well as attorneys and courts, we hope they will gain an appreciation of the distinct relationship

among boundaries, boundary creation, and boundary retracement. When the term *boundary* is referred to in this sense, it is land boundaries that are intended, not property boundaries, in that a surveyor may conduct a property survey but that should be considered as a special-request survey that is specific and not universal in nature. A boundary line may be altered and become a property line boundary by special doctrines that are primarily legal in nature. Estoppel, acquiescence, adverse possession, and agreement are legal doctrines that may relegate deed or surveyed boundary lines to property boundary lines. These are legal dicta and should not be addressed by a registered surveyor. They should be understood, however.

There are two particular decisions that each boundary survey should take notice of: *Kerr v. Fee*¹³ and *Rivers v. Lozeau*.¹⁴ Rather than express the authors' personal views, it is better to quote what the courts expressed in these two decisions. In *Kerr*, the court stated as follows: "To survey land means to ascertain the corners, boundaries and divisions, with distances and directions, and not necessarily to compute areas included in defined boundaries. Knowing these, any competent mathematician can ascertain the areas."

Then in 1989, the Florida Supreme Court, in *Rivers v. Lozeau*, identified the responsibilities of the modern surveyor with these words:

First: "The definition of a legally sufficient real property description is one that can be located on the ground by a surveyor."

Second: "In a retracement survey, not a resurvey, . . . each succeeding surveyor is a following or tracing surveyor, and his sole duty, function and power is to locate on the ground the boundaries corners and boundary lines established by the original survey. . . ."

Principle 11. *When there is a scarcity of written evidence that indicates no lines were created on the ground, any survey conducted subsequently may be held as the first survey.*

Whether a survey is considered as an original survey or a first survey may be a matter of being able to correlate the field evidence as well as the documentary evidence as to time of creation. Some courts have held that when the documentary evidence and the field evidence had to be created at the same time. But on the other hand other judges have determined that if the of field survey was conducted near the time of the documentation that would suffice to meet the requirements.

The object of an original survey is to create boundary lines.
The object of a retracement is to find the evidence (footsteps) of the original surveyor and then to redescribe the footsteps by today's standards.

NOTES

1. William Leybourn, *The Compleat Surveyer* (1657). Reprinted by Walter G. Robillard.
2. John Love, *Geodaesia* (London: 1687). Reprinted by Walter G. Robillard.